

CSEg 1101 Introduction to Computing

Lecture 4 + HW#2



Computer Science and Engineering
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OUTLINE

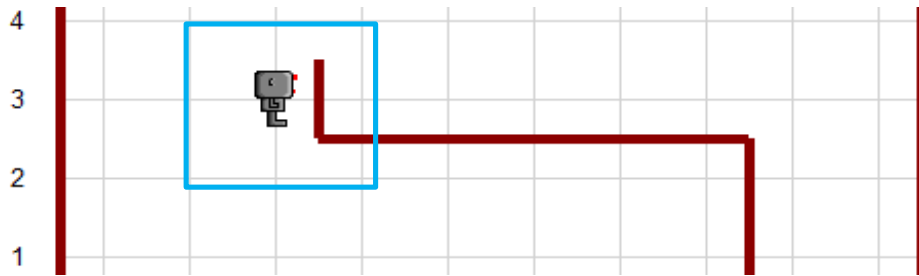
Conditionals

While-loops

Practice

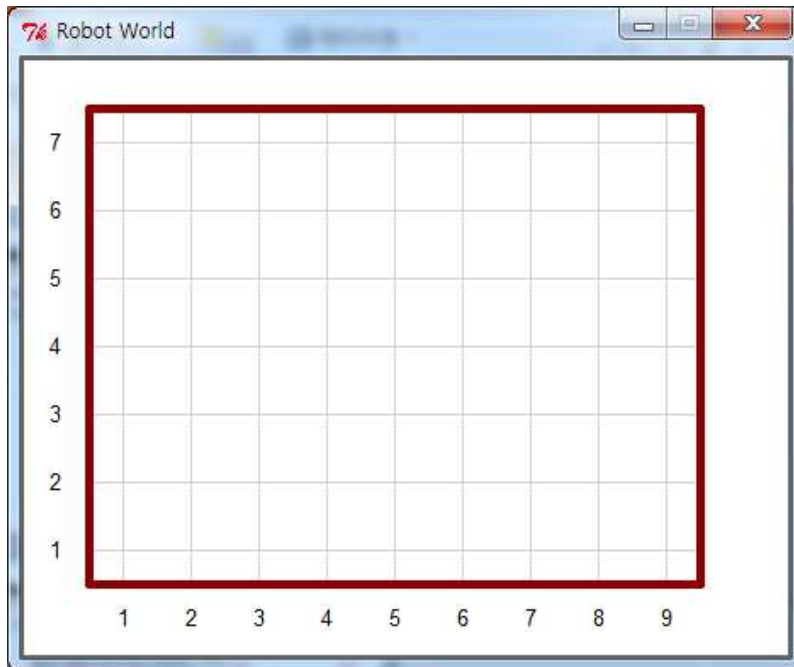
To check if there is a wall on the left, right, or front.

```
hubo.front_is_clear()  False  
hubo.left_is_clear()   True  
hubo.right_is_clear()  True
```



To create a grid

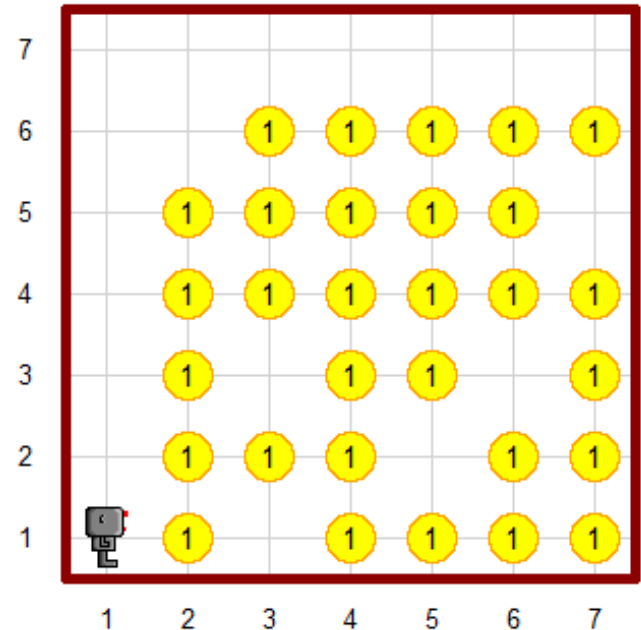
```
create_world(avenues = 9, streets = 7)
```



PRACTICE USING CONDITIONALS

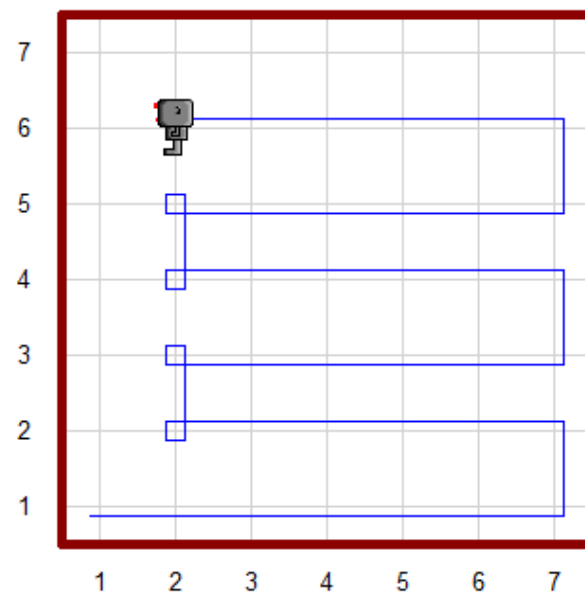
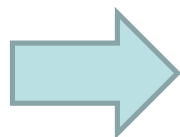
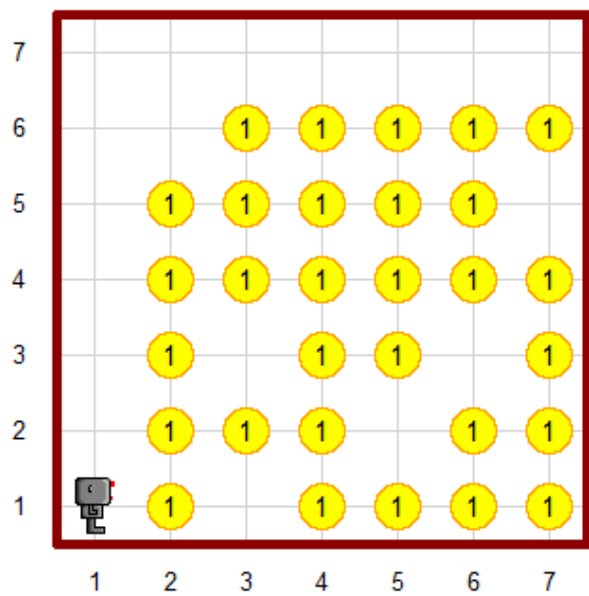
PROBLEM : HARVEST4*

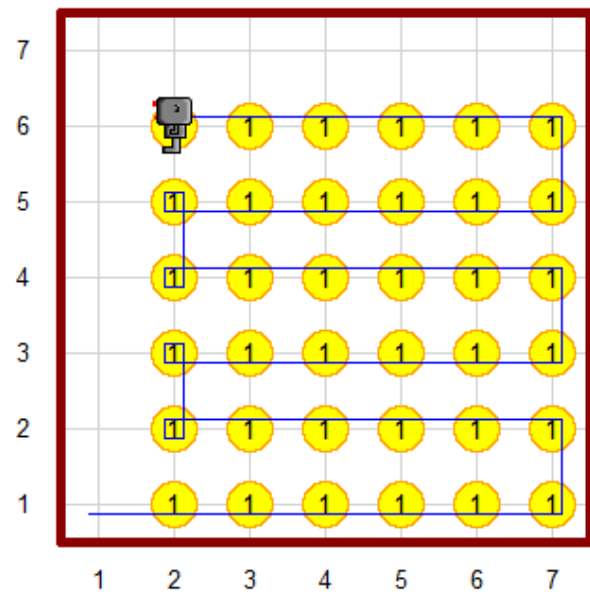
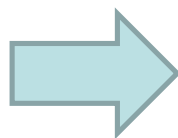
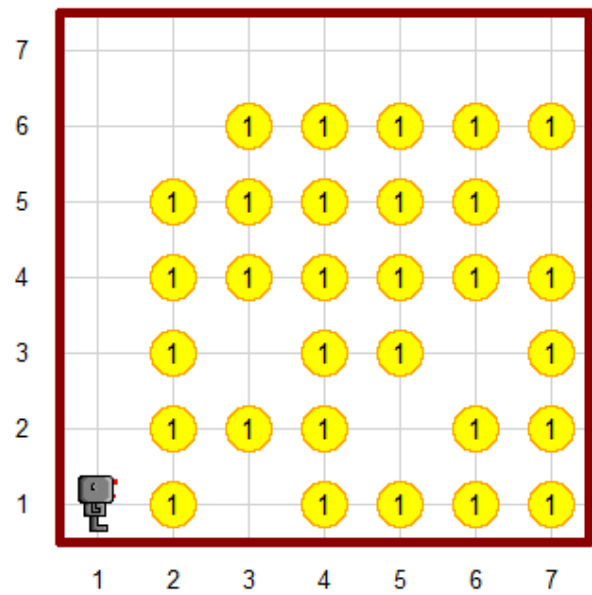
Modify your program for the Harvest2 task so that the program works for the world harvest3.wld. Note that the new program should also automatically work for the original harvest.wld world.



Load the file, harvest3:

```
load_world("./worlds/harvest3.wld")
```





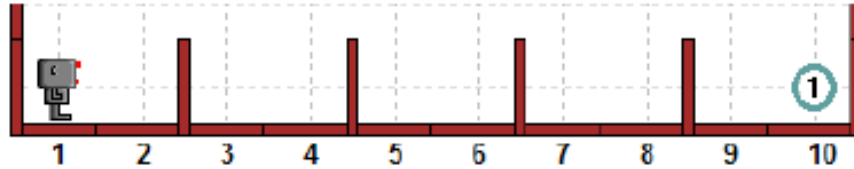
PROBLEM: HURDLES3*

Our robot has become quite good at jumping hurdles. Hubo now enters races of different lengths: short sprints and long races. He knows that he has reached the finish line when he gets to the position with a beeper. Assume that there are no races longer than 20 units.

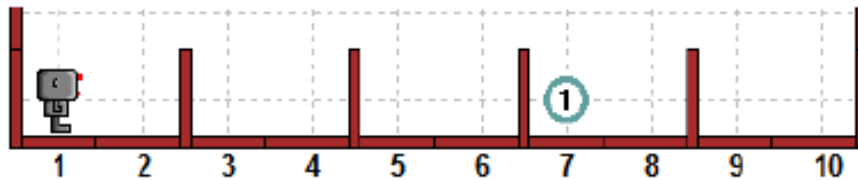
```
for i in range(20):  
    move_jump_or_finish()
```

Use `exit(1)` in the `sys` module to finish a race.

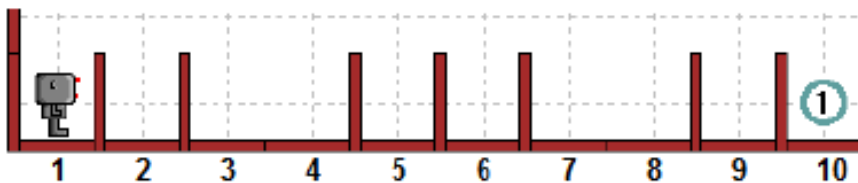
Below, you will find three such race courses; the world files are hurdles1.wld, hurdles2.wld, and hurdles3.wld :



hurdles1.wld



hurdles2.wld



hurdles3.wld

Summary

conditionals: `if`, `if~else`, and `if~elif~else`

iterations

for-loops

while-loops

assignment, e.g., `hubo = Robot()`

functions

You have picked up the main constructs for programming.